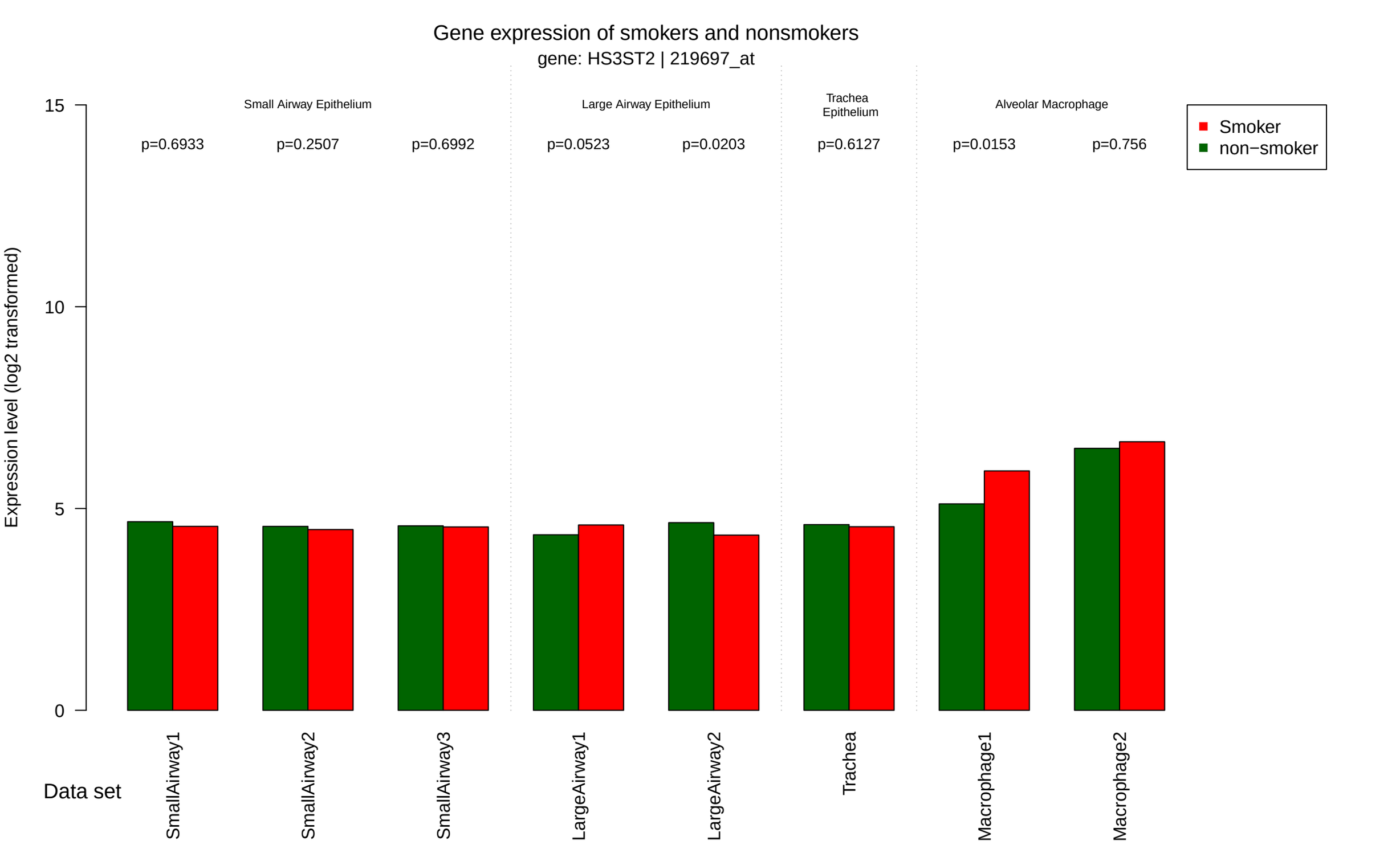
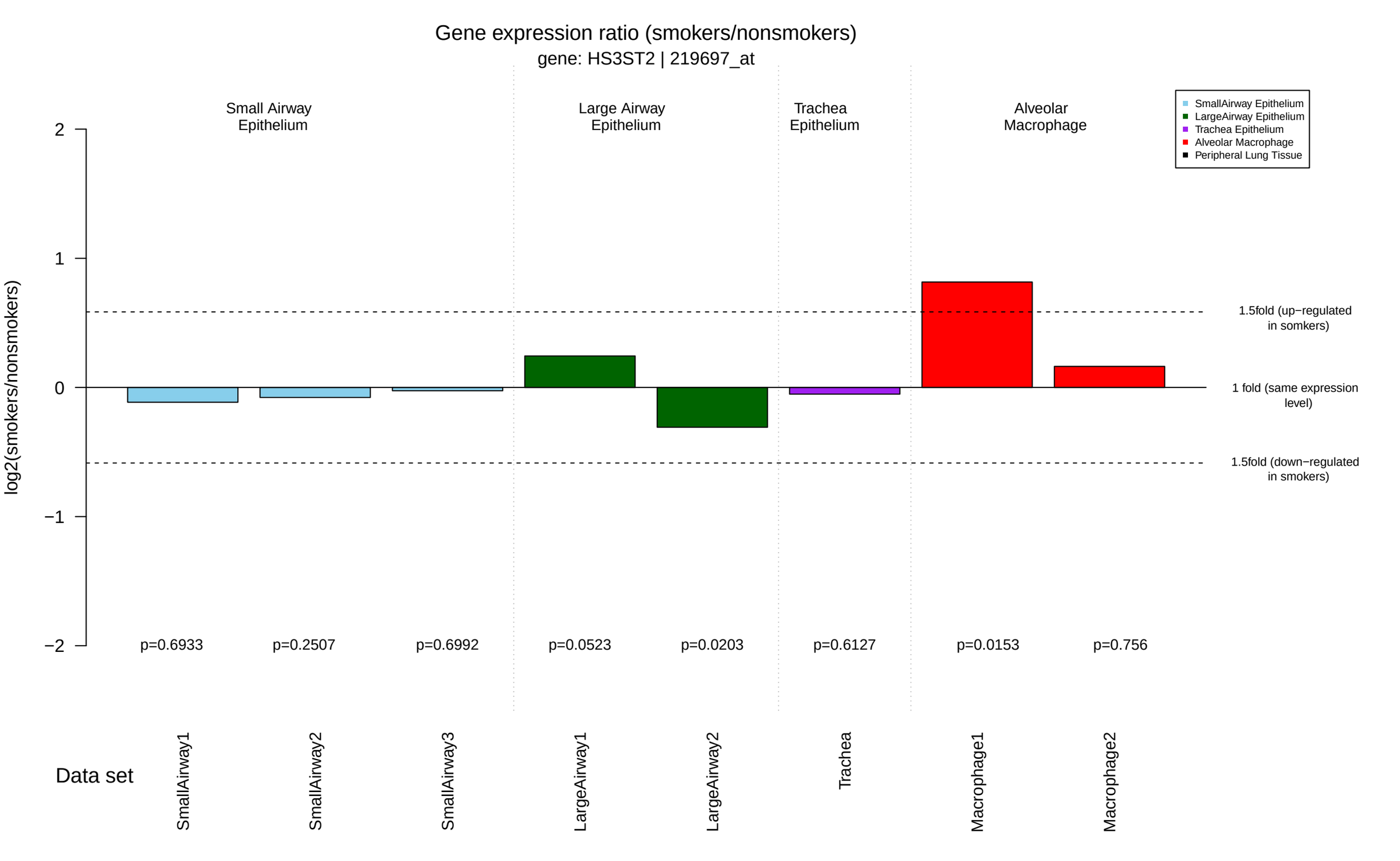
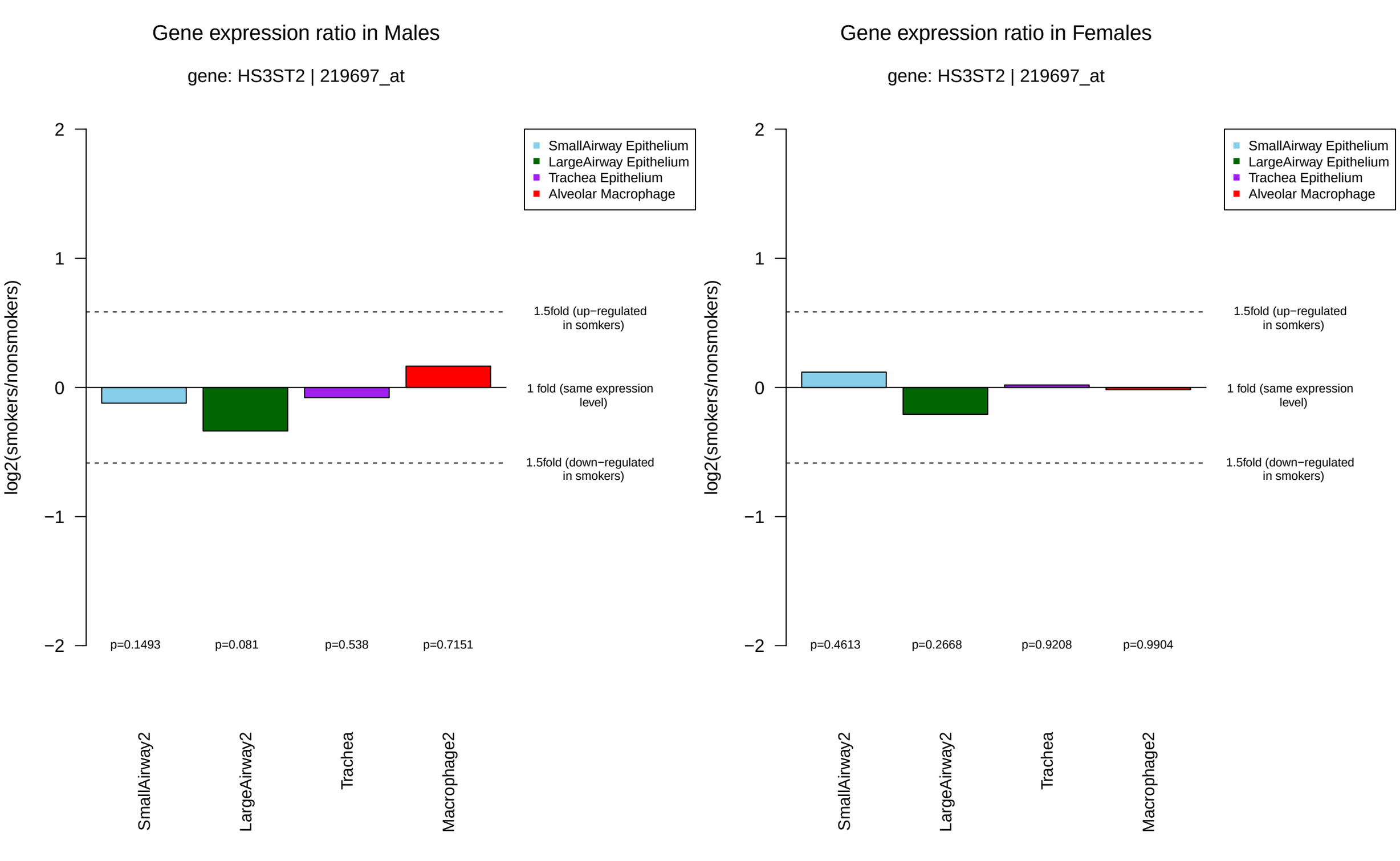
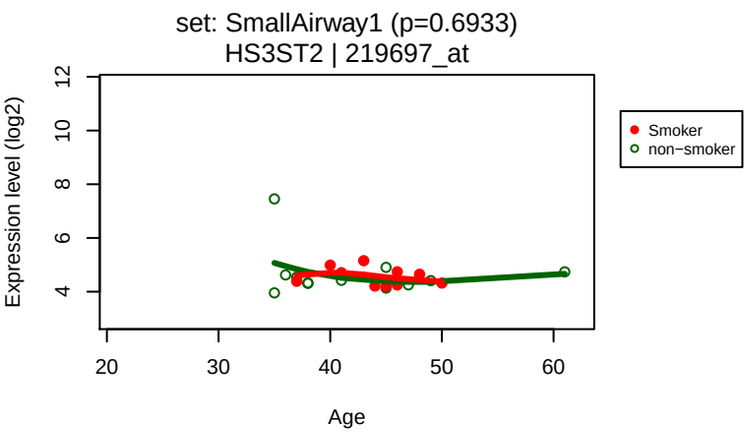
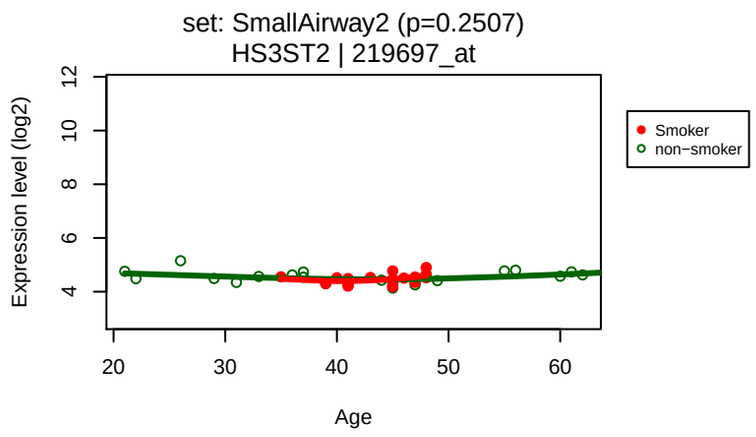
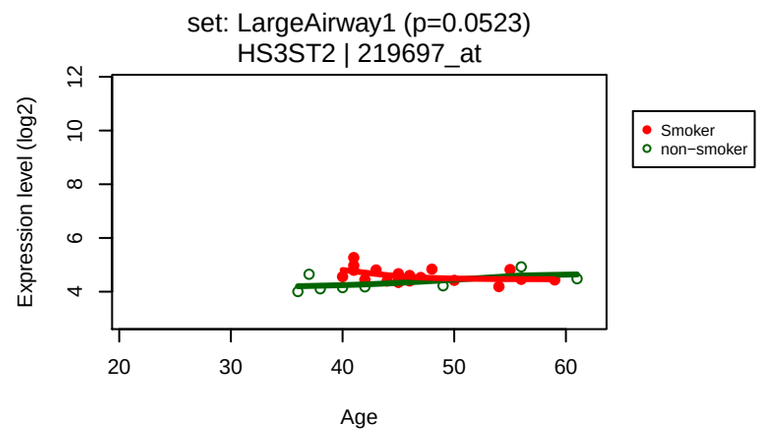

Gene expression of smokers and nonsmokers
gene: HS3ST2 | 219697_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6933
p=0.2507
p=0.6992
p=0.0523
p=0.0203
p=0.6127
p=0.0153
p=0.756
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

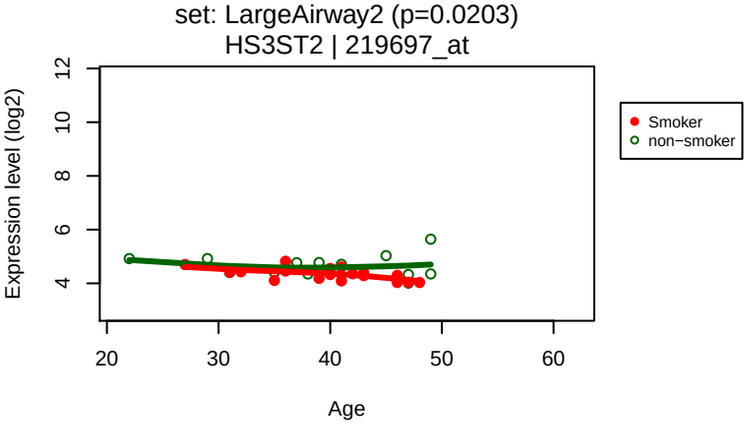

Gene expression ratio (smokers/nonsmokers)
gene: HS3ST2 | 219697_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6933
p=0.2507
p=0.6992
p=0.0523
p=0.0203
p=0.6127
p=0.0153
p=0.756
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

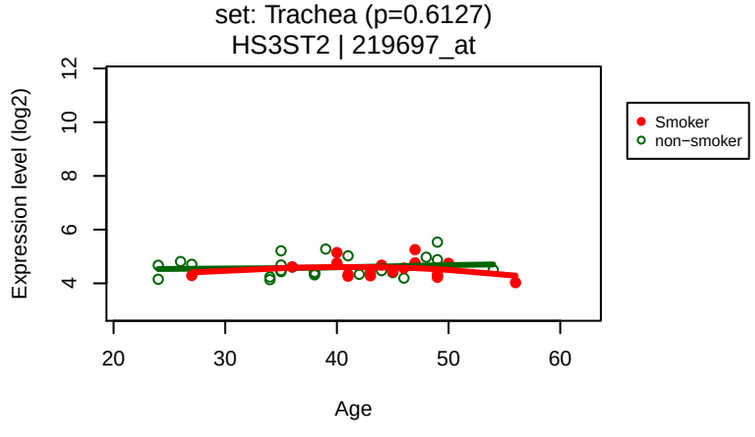

Gene expression ratio in Males
gene: HS3ST2 | 219697_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1493
p=0.081
p=0.538
p=0.7151
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HS3ST2 | 219697_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4613
p=0.2668
p=0.9208
p=0.9904
SmallAirway2
LargeAirway2
Trachea
Macrophage2

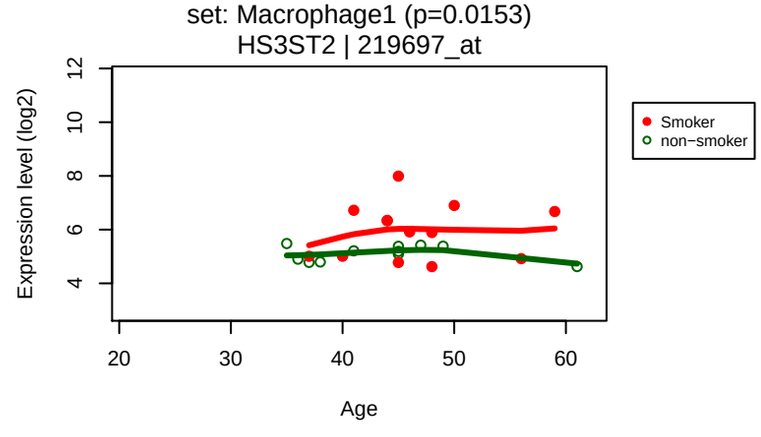

set: SmallAirway1 (p=0.6933)
HS3ST2 | 219697_at
Expression level (log2)
Age
Smoker
non−smoker

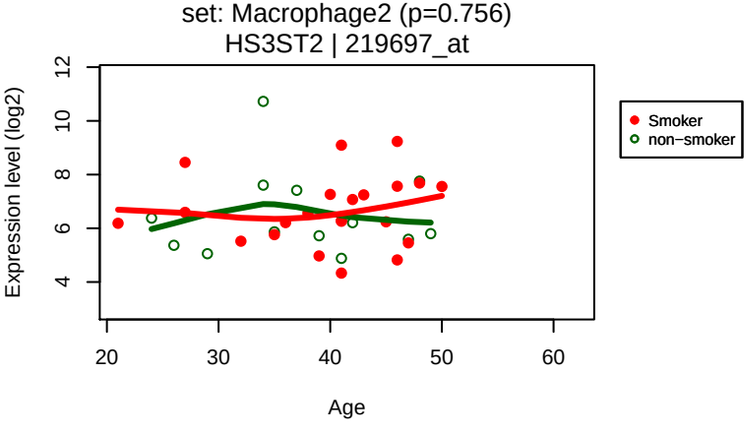

set: SmallAirway2 (p=0.2507)
HS3ST2 | 219697_at
Expression level (log2)
Age
Smoker
non−smoker

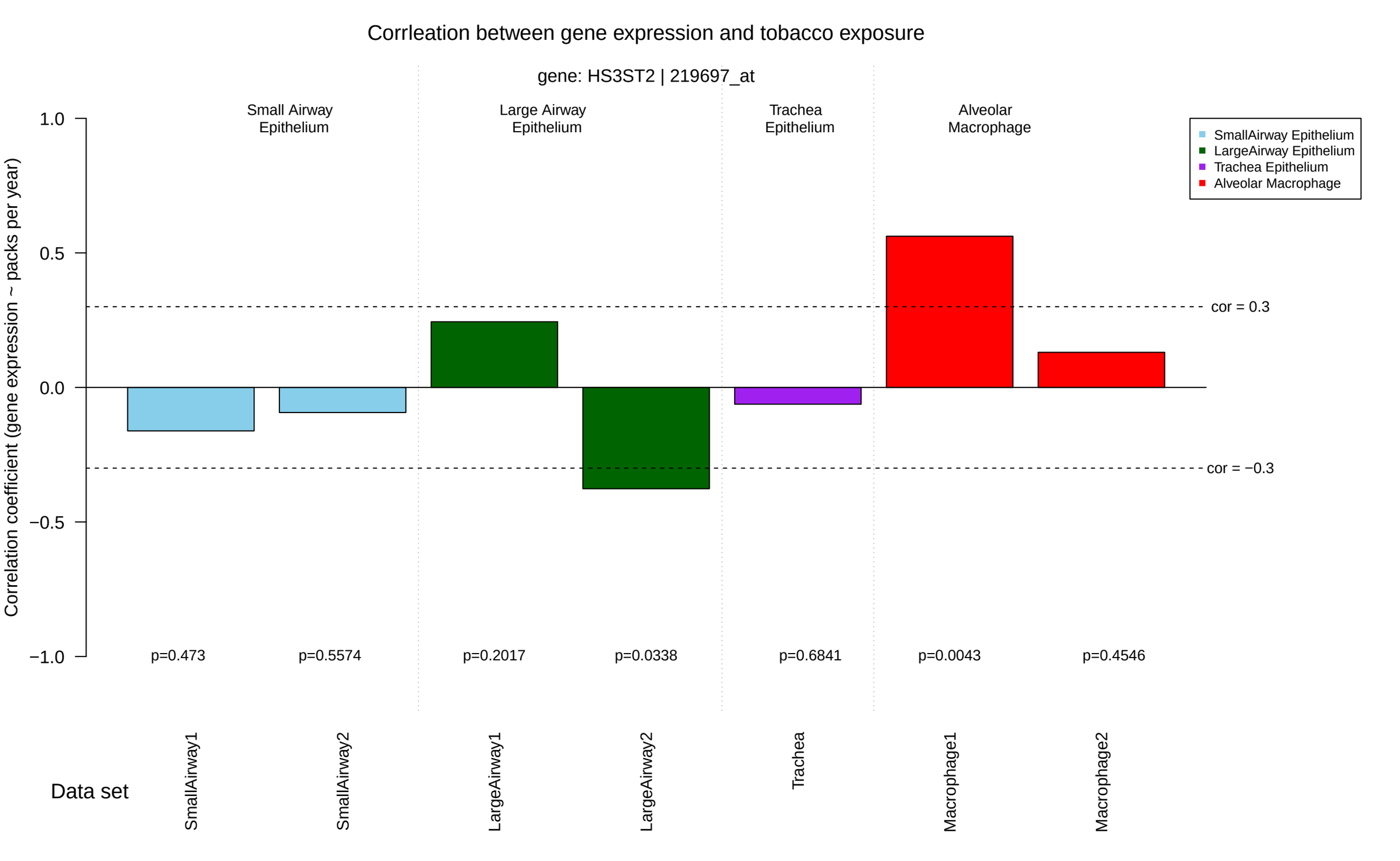

set: LargeAirway1 (p=0.0523)
HS3ST2 | 219697_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0203)
HS3ST2 | 219697_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6127)
HS3ST2 | 219697_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0153)
HS3ST2 | 219697_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.756)
HS3ST2 | 219697_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HS3ST2 | 219697_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.473
p=0.5574
p=0.2017
p=0.0338
p=0.6841
p=0.0043
p=0.4546
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2